Problem Set 4: Multidimensional Distributions, Law of Large Numbers, Central Limit Theorem and Empirical Statistics

(March 15 – April 11, 18:00)

Problem 1 (ZVV)

a) If a 6 tram and and a 10 tram, both heading towards the main train station, arrive at the tram stop "ETH/Universitätsspital" with a temporal separation of less than $\tau = 30s$, one of them will be slowed down and forced to wait on the open track; we refer to this as a "collision".

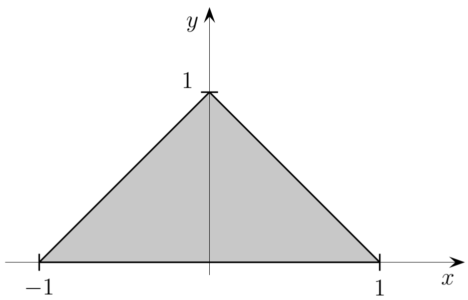
During evening rush hour, trams no longer follow a schedule, but arrive randomly. We assume that two trams (one from line 6, one from line 10) arrive independently and uniformly distributed over the time interval $[0, T]$ ($T = 5$ min). What is the probability of a collision?

Hint: Draw a $[0, T] \times [0, T]$ square where each axis corresponds to the arrival time of a tram. Find the area on the square where a collision occurs.

b) Theo takes the tram every day from "Milchbuck" to "ETH/Universitätsspital". In particular, he is indifferent between taking Trams 9 and 10. He notices that he takes the Tram 10 75% of the time and concludes that Tram 10 is more frequent than Tram 9. What do you think about his conclusion?

Problem 2 (Triangle Density)

The joint density of the two random variables X and Y is constant and equal to c within the gray triangle and 0 outside.



a) Determine the joint density function f of X and Y.

b) Determine the marginal densities.

Hint: Distinguish in the calculation between the regions $\{x < -1\}$; $\{-1 \leq x < 0\}$; $\{0 \leq x < 1\}$ and $\{1 \leq x\}$.

c) Determine $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathrm{Var}(X)$, and $\mathrm{Var}(Y)$.

d) Determine the covariance between X and Y. Are X and Y independent? Justify your answer.

e) Given $X = 0.5$, what is the conditional expectation of Y?

Problem 3 (Covariances)

The marginal densities of a two-dimensional random vector $Z = (X, Y)$ are defined as follows:

$$f_X(x) = \begin{cases} \frac{1}{2} & -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \qquad f_Y(y) = \begin{cases} \frac{3}{4}(2y - y^2) & 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The correlation coefficient ρ_{XY} between X and Y is $\rho_{X,Y} = \sqrt{\frac{1}{3}}$.

a) Calculate the expectation of $6X - 4Y + 2$.

b) Calculate the covariance $\mathrm{Cov}(6X, 4Y)$.

c) Calculate the variance of $6X - 4Y + 2$.

d) Calculate the expectation of $6X^2 - 4Y^2$.

Problem 4 (Batteries)

The expected lifespan μ of a type of battery is unknown and is to be estimated by the arithmetic mean of the lifespans of n independent test batteries of this type. Experience shows that the standard deviation of the lifespan is approximately 5 hours. How large must n be at least so that the absolute difference between the arithmetic mean and μ is at most 1 hour with a probability of at least 99%?

Hint: Use the Central Limit Theorem.

Problem 5 (Central Limit Theorem Simulations)

On http://onlinestatbook.com/stat_sim/sampling_dist/index.html you will find a tool that is very helpful for understanding the Central Limit Theorem.

Let $X_1, \ldots, X_N$ be i.i.d. random variables. As usual, let $\bar{X}_N = \frac{1}{N}\sum_{i=1}^{N} X_i$ be the arithmetic mean.

A short guide: Choose "Begin" in the menu on the left side (to learn about further functionalities, you can also first read the "Instructions" section).

- The top plot shows the distribution of a single X_i (or a histogram of it). This distribution is initially set to "Normal" but can be chosen in a dropdown menu or manually changed by clicking on the plot.
- By default, the number of observations (the "Sample Size") is set to $N = 5$, it can be increased in the dropdown menu below. Clicking on "Animated" now generates an observation of $X_1, \ldots, X_N$ and plots them all in the "Sample Data" plot. Then, the mean $\bar{X}_N$ of this sample is plotted in the "Distribution of Means" plot.
- Another click on "Animated" repeats this process. Repeat this several times to ensure that you have understood the previous point. Clicking on 5, 10,000, or 100,000 repeats the process 5, 10,000, or 100,000 times. However, the observations are no longer plotted in the "Sample Data" plot, instead, for each draw, the arithmetic mean of the N observations is plotted directly in the "Distribution of Means" plot.
- This plot now shows the empirical distribution of $\bar{X}_N$. If the sample size N is large enough, according to the Central Limit Theorem, it should roughly resemble a normal distribution. By checking "Fit normal", you can verify how well this holds.

a) How large must be the sample size N be so that the distribution of $\bar{X}_N$ roughly resembles a normal distribution when X_i is Normal, Uniform or Skewed?

b) Pick any of the three distributions and any N and simulate 100,000 samples of $\bar{X}_N$. What is the standard deviation of the means? Is this what you expected?

c) This is intended to be a visualization of the Central Limit Theorem. However, one can also visualize the Law of Large Numbers if viewed correctly. How?

Problem 6 (Empirical Statistics)

Consider the following 10 data points $x_1, \ldots, x_{10}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
0.42	0.45	3.11	1.33	1.68	0.09	2.02	5.35	0.71	2.29

a) Give the values of the ordered data $x_{(1)}, \ldots, x_{(10)}$.

b) Find the arithmetic mean and the empirical standard deviation of the data points $x_1, \ldots, x_{10}$.

c) Find the median, lower quartile and the upper quartile of the data points $x_1, \ldots, x_{10}$.

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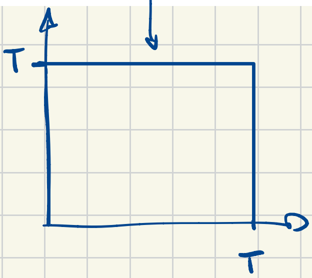
a)



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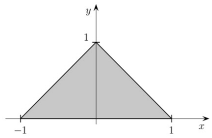

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d) Determine the covariance between X and Y. Are X and Y independent? Justify your answer.

e) Given $X = 0.5$, what is the conditional expectation of Y?

a) same Analysis II

b) $f_X = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$

↳ same for f_Y

c) formulas from lecture

d)

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↳ look at cheatsheet

→ properties of $\mathbb{E}(X)$, $\mathrm{Cov}(X)$, $\mathrm{Var}(X)$

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Hint: Use the Central Limit Theorem.

$\bar{L}_n = \frac{1}{n}\sum_{i=1}^{n} L_i$ with $E(\bar{L}_n) = \mu$ ← law of large numbers

↳ we look for $P\left[|\bar{L}_n - \mu| \leq 1\right] \geq 0.99$

→ CLT: $\bar{L}_n$ is normally distributed

Problem 5)

good for intuition

else: 3Blue1Brown vid of CLT

Problem 6

as hinted in exercise

remember: $P(A \cap B) = P(A) \cdot P(B)$

$\Rightarrow$ independent